

AUTOMATIC SMART COMPOST BIN

A MINI PROJECT REPORT

Submitted to

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bonafide students of Shri Madhwa Vadiraja Institute of Technology and Management, in completion of “**Mini Project**” in Electronics & Communication Engineering of Visvesvaraya Technological University, Belagavi during the year 2025-2026. It is certified that all the corrections / suggestions indicated during Internal Assessment have been incorporated in the report. The report has been approved as it satisfies the academic requirements in respect of Mini Project Work prescribed for the said degree.

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ABSTRACT

The project presents the design and development of an Automatic Compost Bin aimed at improving organic waste management in small-scale institutional and domestic environments. The system integrates both mechanical and electronic components to automate key composting processes such as waste mixing, aeration and environmental monitoring. A motorized blade mechanism ensures uniform mixing, while sensor-based feedback enables real-time control of temperature and moisture conditions essential for efficient composting. The system is further enhanced with IoT connectivity through the Blynk platform, allowing users to remotely monitor and manage the composting process via a smartphone application. By minimizing manual intervention and optimizing composting parameters, the proposed system significantly reduces composting time and ensures the production of high-quality, nutrient-rich compost. Compact, user friendly and sustainable. This solution offers a practical approach to converting organic waste into valuable compost for hostels, homes and community kitchens.

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LIST OF ACRONYMS AND ABBREVIATIONS

IOT	Internet Of Things
CAD	Computer Aided Design
DHT22	Digital Humidity And Temperature Sensor
NPK	Nitrogen, Phosphorus, Potassium Sensor
DS18B20	Digital Sensor 1-Wire 1